“An Intelligent Performance Evaluation System Integrating SMART and Deep Learning for Human Resource Analytics”

A Project Abstract

submitted to

MOHAN BABU UNIVERSITY, TIRUPATI

In Partial Fulfillment of the Requirements for the Award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

BY

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2022-2026

ABSTRACT

Employee performance evaluation is a critical factor in achieving organizational goals, yet traditional appraisal methods often suffer from subjectivity, bias, and inefficiency. This project proposes an intelligent performance estimation system that integrates the Simple Multi-Attribute Rating Technique (SMART) for structured, multi-criteria evaluation with Deep Learning models for analyzing behavioral aspects such as teamwork, adaptability, communication skills, and problem-solving ability.

The system collects both quantitative metrics (task completion rate, quality of output, punctuality) and qualitative inputs (peer feedback, 360-degree reviews), assigns criterion weights through SMART, and processes the data using predictive models like LSTM, CNN, and Transformers to improve accuracy. Visualization tools such as Matplotlib, Seaborn, and Plotly present the results in interactive dashboards for HR and management.

By combining SMART's decision-making framework with AI-driven behavioral analytics, the proposed system enables faster, transparent, and unbiased performance evaluations. This approach enhances employee engagement, supports data-driven HR decision-making, and adapts to diverse organizational needs, thereby promoting both fairness and productivity.

Details of Base Paper:

Luo, Z., & Lu, K. (2025). Application research and effectiveness evaluation mechanism of hybrid intelligent algorithm integrating cognitive computing and deep learning for dynamically adjusting employee performance evaluation in multi-scale organizational networks. Discover Artificial Intelligence, 5, Article 45. https://link.springer.com/article/10.1007/s44163-025-00285-x

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